Nonlinear Practice 4

Module 8 - Nonlinear Pharmacokinetics · Daily practice sheet · No date

Name _____ Date _____

A 68-kg, 28-year-old female is receiving phenytoin for the treatment of seizures. When this patient was taking a daily oral dose of 300 mg, her steady state plasma concentration was 9 mg/L. The dose was increased to 400 mg which resulted in a steady-state plasma concentration of 18 mg/L. Assume 100% bioavailability and an apparent volume of distribution that is equal to 70% of body weight.

a Calculate KM

b Calculate Vmax

c Calculate the half-life and clearance in this patient at steady-state while she is taking 300 mg/day. What is the half-life?

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d Calculate the half-life and clearance in this patient at steady-state while she is taking 400 mg/day. What is the half-life?

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e What is the daily phenytoin dose that should achieve a steady-state phenytoin plasma concentration of 15 mg/L in this patient?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

a

Her key

Answer: 9 mg/L

WHAT THE STEM GIVES

R1	300 mg	the first daily dose
C_1	9 mg/L	steady state on 300 mg
R2	400 mg	the increased daily dose
C_2	18 mg/L	steady state on 400 mg
weight	68-kg	*not needed for KM*

THE WORKING, ONE LINE AT A TIME

SETUP

$$KM = \frac{R2 - R1}{(R1/C_1) - (R2/C_2)}$$, with R1 = 300 mg/day, C_1 = 9 mg/L, R2 = 400 mg/day, C_2 = 18 mg/L

Two daily doses in the same patient and the steady-state concentration each produced: the direct method. The 68 kg weight and the 70% volume are needed only for the half-life in parts c and d.

ALGEBRA

$$KM = \frac{400 - 300}{(300/9) - (400/18)} = \frac{100}{33.333 - 22.222} = \frac{100}{11.111} = 9.000 \text{ mg/L}$$

Each dose over its own concentration, the earlier pair first so that the denominator is positive. The mg/day cancel and leave mg/L, a concentration.

ROUND

KM = 9 mg/L

9.000 mg/L exactly, because 33.333 − 22.222 is one ninth of 100. Here KM equals the 9 mg/L measured on 300 mg/day.

> **DOES THE NUMBER MAKE SENSE?**
>
> A one-third rise in dose doubled the concentration, from 9 to 18 mg/L, so elimination is saturable; KM is a concentration: 9 mg/L.

b

Her key

Answer: 600 mg/day

WHAT THE STEM GIVES

KM	9 mg/L	from part a, added to C_{ss}
R2	400 mg	the dose used
C_2	18 mg/L	its own steady state
R1, C_1	300 mg, 9 mg/L	the check pair

THE WORKING, ONE LINE AT A TIME

SETUP

$$\text{Vmax} = \frac{R(KM + C_{ss})}{C_{ss}}$$

The dose-rate line solved for Vmax: multiply both sides by (KM + C_{ss}) and divide by C_{ss}, using one dose and the steady-state concentration it produced.

ALGEBRA

$$\text{Vmax} = \frac{(400\ \text{mg/day})(18 + 9)\ \text{mg/L}}{18\ \text{mg/L}} = \frac{10\ 800}{18} = 600\ \text{mg/day}$$

Her sheet uses the second pair, 400 mg/day with its own 18 mg/L. mg/L cancels and Vmax keeps the units of the dose, mg/day.

ALGEBRA

$$\text{Check: } \frac{(300\ \text{mg/day})(9 + 9)\ \text{mg/L}}{9\ \text{mg/L}} = \frac{5400}{9} = 600\ \text{mg/day}$$

The first pair gives the same 600 mg/day, as it must for one patient with one Vmax. Pairing 300 mg/day with 18 mg/L instead gives 450 mg/day, a mismatched pair.

> **DOES THE NUMBER MAKE SENSE?**
> Vmax must exceed 400 mg/day, the higher dose, because that dose reached a steady state: 600 mg/day.

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c Her key

Answer: 0.989 days

WHAT THE STEM GIVES

weight	68-kg	$V_D = 0.7 \times 68 = 47.6$ L
V_D	70% of body weight	0.7 L/kg
C_p	9 mg/L	the steady state on 300 mg/day
KM	9 mg/L	from part a
Vmax	600 mg/day	from part b

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 0.7\ \text{L/kg} \times 68\ \text{kg} = 47.6\ \text{L}$$

"70% of body weight" is read as 0.7 L/kg, taking 1 L as 1 kg, so the volume is 0.7 times the 68 kg weight, in litres, which the half-life line needs.

SETUP

$$t½ = V_D\left(\frac{0.693}{\text{Vmax}}\right)(KM + C_p), \text{ at } C_p = 9\ \text{mg/L}$$

On 300 mg/day the patient is at the steady state that dose produced, 9 mg/L, and the half-life of a saturable drug is calculated at its concentration.

ALGEBRA

$$t½ = 47.6\ \text{L} \times \frac{0.693}{600\ \text{mg/day}} \times (9 + 9)\ \text{mg/L} = 0.9896\ \text{days}$$

L × mg/L gives mg, which cancels the mg of mg/day and leaves days. Leaving the volume as 0.7 gives 0.015, a result per kilogram, not a time.

ROUND

$$t½ = 0.989\ \text{days}$$

Her sheet writes 0.989 day. 0.9896 days to three decimal places is 0.990, and both are accepted. On 400 mg/day the half-life is longer, because KM + C_p is larger.

DOES THE NUMBER MAKE SENSE?

Vmax is in mg/day, so the half-life comes out in days: 0.9896 days, just under one day.

Her sheet writes 0.989 day; the arithmetic gives 0.9896 day, which is 0.990 to three decimal places. Both are accepted.

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c Her key

Answer: 33.33 L/day

WHAT THE STEM GIVES

C_p	9 mg/L	the steady state on 300 mg/day
KM	9 mg/L	from part a
Vmax	600 mg/day	from part b
weight	68-kg	*not needed for clearance*

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = \frac{\text{Vmax}}{\text{KM} + C_p}, \text{ at } C_p = 9\ \text{mg/L}$$

On 300 mg/day the patient is at 9 mg/L, and the clearance of a saturable drug is calculated at its concentration, which sits in the denominator.

ALGEBRA

$$Cl_T = \frac{600\ \text{mg/day}}{(9 + 9)\ \text{mg/L}} = \frac{600}{18} = 33.33\ \text{L/day}$$

mg/day over mg/L leaves L/day, a volume of plasma cleared per unit time. The weight and the volume of distribution are not needed for clearance.

ROUND

$$Cl_T = 33.33\ \text{L/day}$$

To two decimal places, as her sheet gives it. On 400 mg/day the clearance is lower, because the larger C_p makes the denominator larger.

DOES THE NUMBER MAKE SENSE?

At 9 mg/L, C_p equals KM, so KM + C_p is twice KM and clearance is half of Vmax ÷ KM: 33.33 L/day.

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d Her key

Answer: 1.48 days

WHAT THE STEM GIVES

weight	68-kg	$V_D = 0.7 \times 68 = 47.6$ L
V_D	70% of body weight	0.7 L/kg
C_p	18 mg/L	the steady state on 400 mg/day
KM	9 mg/L	from part a
Vmax	600 mg/day	from part b

THE WORKING, ONE LINE AT A TIME

UNIT

$$V_D = 0.7 \text{ L/kg} \times 68 \text{ kg} = 47.6 \text{ L}$$

"70% of body weight" is read as 0.7 L/kg, so the volume is 0.7 times the 68 kg weight, in litres. It does not change with the dose; only KM + C_p does.

SETUP

$$t½ = V_D\left(\frac{0.693}{Vmax}\right)(KM + C_p), \text{ at } C_p = 18 \text{ mg/L}$$

On 400 mg/day the patient is at the steady state that dose produced, 18 mg/L, so the half-life line is evaluated at that concentration.

ALGEBRA

$$t½ = 47.6 \text{ L} \times \frac{0.693}{600 \text{ mg/day}} \times (9 + 18) \text{ mg/L} = 1.484 \text{ days}$$

The same line as in part c with C_p = 18 mg/L. KM + C_p goes from 18 to 27 mg/L, so the half-life grows by the same ratio, one and a half times.

ROUND

$$t½ = 1.48 \text{ days}$$

1.484 days to two decimal places, her sheet's value: longer than the 0.989 day on 300 mg/day, because the half-life of a nonlinear drug lengthens as the concentration rises.

> **DOES THE NUMBER MAKE SENSE?**
> The half-life at 18 mg/L must be longer than at 9 mg/L, because KM + C_p is larger: 1.484 days.

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d Her key

Answer: 22.22 L/day

WHAT THE STEM GIVES

C_p	18 mg/L	the steady state on 400 mg/day
KM	9 mg/L	from part a
Vmax	600 mg/day	from part b

THE WORKING, ONE LINE AT A TIME

SETUP

$$Cl_T = \frac{Vmax}{KM + C_p}, \text{ at } C_p = 18 \text{ mg/L}$$

The same clearance line as in part c, now at the 18 mg/L that 400 mg/day produced, because the clearance of a saturable drug changes with concentration.

ALGEBRA

$$Cl_T = \frac{600 \text{ mg/day}}{(9 + 18) \text{ mg/L}} = \frac{600}{27} = 22.22 \text{ L/day}$$

mg/day over mg/L leaves L/day. The larger concentration makes the denominator larger, so the clearance is smaller than the 33.33 L/day on 300 mg/day.

ROUND

$$Cl_T = 22.22 \text{ L/day}$$

To two decimal places, as her sheet gives it. Clearance falls and half-life rises together as the dose rises, because both depend on KM + C_p.

DOES THE NUMBER MAKE SENSE?

Clearance at 18 mg/L must be below its value at 9 mg/L, because C_p is in the denominator: 22.22 L/day.

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e

Her key

Answer: 375 mg/day

WHAT THE STEM GIVES

KM	9 mg/L	from part a
Vmax	600 mg/day	from part b
C_{ss}	15 mg/L	the target

THE WORKING, ONE LINE AT A TIME

SETUP

$$R = \frac{Vmax \times C_{ss}}{KM + C_{ss}}$$

The patient's own KM and Vmax, from parts a and b, with the target of 15 mg/L as C_{ss}, in the nonlinear dose-rate line. Vmax is in mg/day, so no weight is needed.

ALGEBRA

$$R = \frac{(600 \text{ mg/day})(15 \text{ mg/L})}{(9 + 15) \text{ mg/L}} = \frac{9000}{24} = 375 \text{ mg/day}$$

mg/L cancels and leaves mg/day. 15 mg/L lies between the 9 and 18 mg/L measured, so the dose lies between the 300 and 400 mg/day she took.

ROUND

$$R = 375 \text{ mg/day}$$

Exactly 375 mg/day, her sheet's value; this stem asks for no rounding. It is below the 600 mg/day Vmax, as every dose rate that reaches a steady state must be.

DOES THE NUMBER MAKE SENSE?

15 mg/L lies between 9 and 18 mg/L, so the dose lies between 300 and 400 mg/day: 375 mg/day.
